MSC THESIS, MEETING WITH THE DIRECTOR, OCTOBER 2026

Governing AI-Agentic Development

A Spec-Driven Neuro-Symbolic Architectural Firewall

Results to date on branch v1.2e, merge commit b703468
Report of 2026-10-12, updated with the director-report-final branch

THE PROBLEM

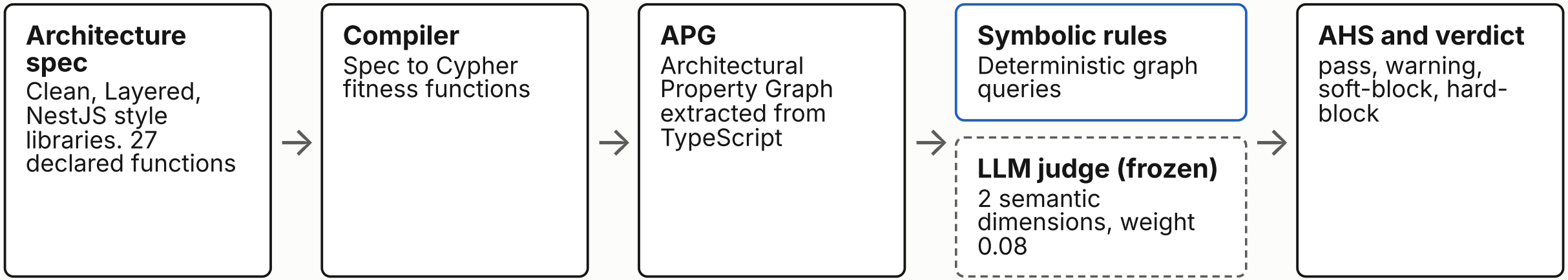
AI coding agents write code faster than anyone can review its architecture

- Generated code compiles and passes tests, yet can quietly break layer rules, add cycles and couple modules.
- A plain "review this against the architecture" prompt is not repeatable and cannot be audited.

> **Research question**
> Can an architecture spec be compiled into executable checks, backed by a frozen LLM judge, so that architectural conformance of agent-written code is measured deterministically, fast and with an audit trail?

WHAT WAS BUILT

DaedalusArch: spec to graph queries to a score



Pre-registration, hash-checked: analysis plan, operator catalogue, matching rule, judge rubric and corpus are frozen before any experiment runs.

OBJECTIVES

SO1 to SO5: all delivered or run, none without a caveat

Objective	Promise	Status on v1.2e
SO1	Spec language and style libraries	Done, Hexagonal not built Clean, Layered, NestJS: 27/27 templates each
SO2	APG coverage, latency, ablation	Done Latency gate passed, FLOWS_TO limitation declared
SO3	Rules plus neural judge, AHS	Partly done Judge agrees with panel, inert at 8%, detection unmeasured
SO4	P/R/F1 on a seeded golden set (80 to 120)	Done with a shortfall N = 75 strict, 73 corrected
SO5	LLM × spec-quality study (E1)	Run 72 cells, 65 judged, P4 labelled

METHOD RIGOUR

Registered first, post-hoc changes declared, results reproduced by an outside audit

v25

Pre-registration version, 85 hash-checked artefacts. The runner refuses a plan whose registered files changed.

11

Post-hoc changes declared in ADRs. Each is reported beside the registered reading and replaces none.

310

Result tables and files reproduced byte-identical by an independent skeptical audit. 0 differ, 22 not reproducible.

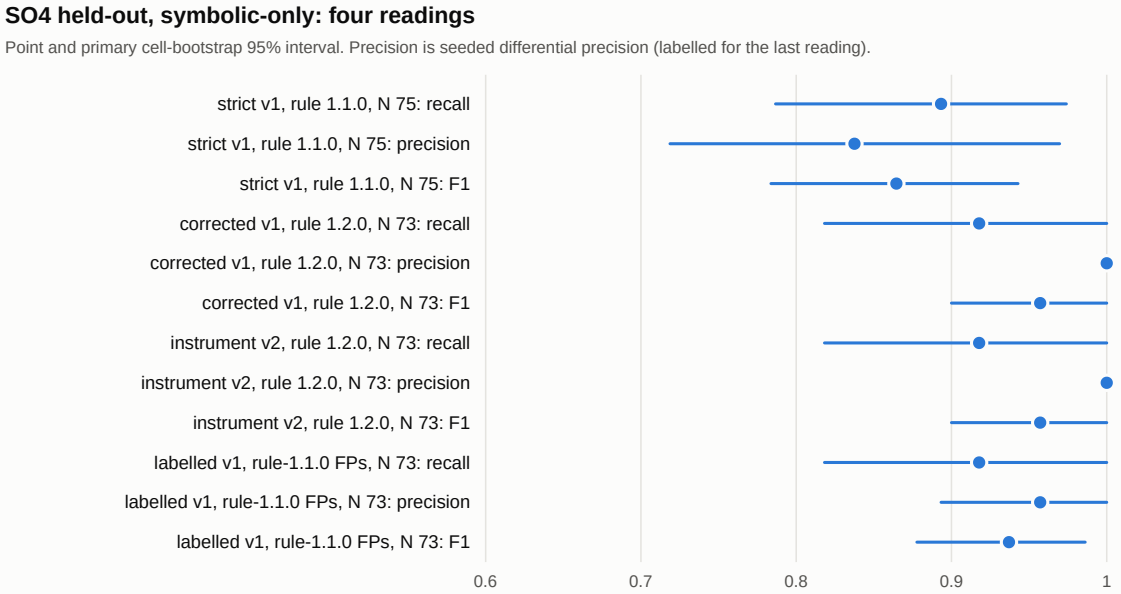
A withdrawn table (labelled SO4 at N = 75) was caught by the audit and replaced under ADR-033.

SO4: SEEDED DETECTION

F1 0.937 on held-out code, from 73 instances against a floor of 80

Reading	N	Recall	Precision	F1
Strict registered	75	0.893 0.787 to 0.974	0.838 0.719 to 0.970	0.865 0.784 to 0.943
Labelled post-hoc	73	0.918 0.818 to 1.000	0.957 0.893 to 1.000	**0.937** 0.878 to 0.986

95% cell-bootstrap intervals. The 6 misses are all dynamic `import()`, outside extractor coverage.



N is 5 below the floor (strict) and 7 below (corrected). Recall is construct sensitivity, not detection of natural violations.

SO4: REAL CODE

Precision on real findings is a range, 0.33 to 0.79

How the 8 `uncertain` items count	Status	Precision (95% CI)
Counted as not real	registered	**0.328** (0.132 to 0.609)
Excluded	post-hoc	0.608 (0.351 to 0.816)
Counted as real	post-hoc	0.789 (0.500 to 0.933)

- 24 labelled items, effective n 11.5. Every reading is below seeded precision.
- The rule for `uncertain` moves the estimate more than the sampling error does.
- Quote the registered 0.33 first. Only the author audit or a second labeller can narrow the range.

SO2

Fast and complete, but the data-flow edge is empty on real code

1.000

Parse coverage on all four real core bases

5.3 s

Full evaluation of the largest base (241 files). The cycle query budget is 30 s.

0

FLOWS_TO edges on the four real bases. A declared limitation (ADR-023).

ABLATION: FULL GRAPH VS AST-ONLY

- Without type-checker edges, FF-P02 (dependency inversion) goes blind on 3 of 4 bases, and FF-P04 on 2.
- The ablation cannot show what FLOWS_TO adds, because there are no such edges to remove.

SO3: THE LLM JUDGE

It agrees with a panel and survives a model swap, but it changes nothing and its detection is unmeasured

98%

Agrees with the panel
0.982 on E1 (n = 36, 95% CI 0.949 to 1.000, AC1 0.982). Both errors were lenient.

0

Robust to a model swap
Opus vs Sonnet: agreement 0.940, kappa 0.766, 0 verdict changes.

8%

Inert by design
At weight 0.08 it can lower the score by at most 0.08. 0 verdict changes in 9 E7 projects and 65 E1 cells.

?

Detection not measured
The panel failed only 2 of 36 items. The planned catch test placed 0 seeds.

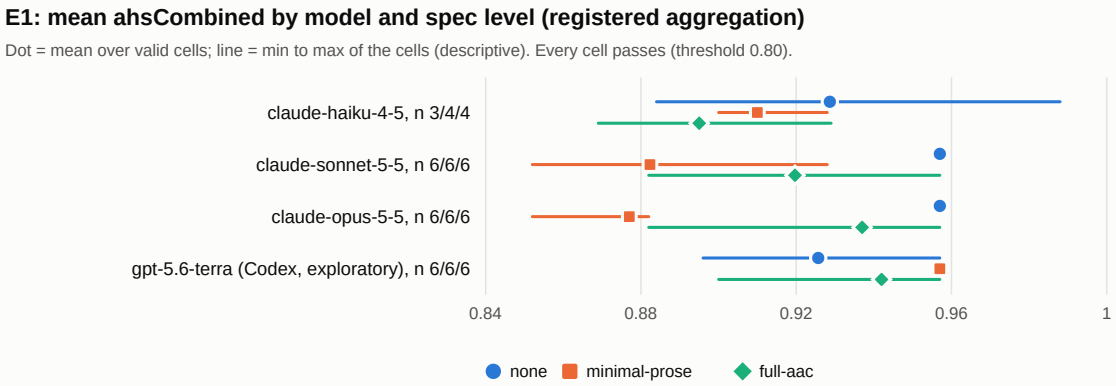
Kappa is 0 on E1 because the judge passed every item (prevalence), so AC1 is the figure to quote. The panel is one Gemini model, not a human.

SO5 (E1): LLM × SPEC LEVEL

The spec level changes the structure the models build; model size does not matter

Spec level	Mean AHS	Files
none	**0.951**	9.1
vague prose	0.887	27.4
precise spec (full-aac)	0.920	29.6

- Spec level: Holm p = 0.0003. Model: p = 0.438.
- At equal size, the precise spec beats vague prose: +0.033, Cliff's delta 0.52 (0.47 size-adjusted).
- No spec gives small, flat projects where two rules rarely fire. Do not claim specs make architecture worse.



COMPARISONS

Equal to dependency-cruiser on import rules; a plain LLM review is not repeatable

VS DEPENDENCY-CRUISER

Seeds	DaedalusArch	dep-cruiser
In coverage (37)	37	28 (36 post-hoc fix)
All seeds (43)	37	34 (42)

- It finds the 6 dynamic-import seeds that we miss.
- Real fix commits: both flag the same 2 of 5.

VS PLAIN LLM REVIEW (PILOT)

Measure	LLM review
Seeds found	9/10 (firewall 5/5)
Run-to-run Jaccard	0.727
Firewall keys covered	26.5%
Cost per review	5.92 credits, 3.2 min

Exploratory: 5 seeds, 25 of 42 reviews, one model.

> The case for the firewall is determinism, exhaustiveness, speed and auditability. It is not better seed detection.

HONEST LIMITATIONS

What the results do not support yet

- **Circularity.** One author wrote the operators, rules and baseline translation. SO4 recall is construct sensitivity.
- **Narrow corpus.** 5 of 7 held-out bases are NestJS. TypeScript back ends of 35 to 819 files only.
- **One labeller model.** The only human check, the 30-item author audit, is still pending.
- **Floor missed.** N = 75 (73) against 80. Real-code precision is 0.33 to 0.79.
- **Judge.** No effect on verdicts, detection unmeasured, no Hexagonal library, dynamic `import()` invisible.
- **E1.** Two tasks, one vendor in the confirmatory tests, every cell passes.

DECISIONS NEEDED FROM THE DIRECTOR

Five calls to make today

1. **Paper order.** E1 empirical paper, tool paper, or LLM-judge lessons paper first, and the venues.
2. **Grok arm.** Add a third vendor to E1, or leave it as future work (recommended).
3. **Judge anti-pattern test.** Approve the redesign (seed shapes surveyed from real code), or state that judge detection is unmeasured (recommended).
4. **Firewall in the loop.** The MCP study where the agent calls the firewall while generating: thesis scope or future work (recommended).
5. **Publication rules.** Registered readings first, post-hoc labelled, exploratory marked. Confirm this for all three papers.

Also open: how to frame the reversed E1 effect, where the Codex arm goes, how to word the P4 result, the 30-item author audit.

NEXT STEPS

Close the evidence gaps, then write

- Complete the 30-item author audit, so that the panel-vs-audit row is not empty.
- Adopt the E1 framing: registered reversed effect first, then the size analysis, then the size-matched contrast.
- Register the proposed API-robustness experiments before running any of them.
- Housekeeping: close threat TV-84, and add a threats entry for the E1 usage-limit misclassification.
- Start the thesis draft and the first paper once the order is chosen.